

Bangkok Airbnb Marketplace Market Intelligence Analysis

Executive Assessment & Statistical Report

Dataset Source: Inside Airbnb (Verified Scraping Data)

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Executive Summary

This report presents a data-based market intelligence examination for the Bangkok Airbnb marketplace through the use of the Inside Airbnb dataset. The foundation of the majority of our engineering effort was to create a scalable, non-parametric end-to-end processing pipeline for handling large amounts of incoming data and for addressing key data quality issues such as complete nulls for pricing.

The statistical validity of the determination was achieved through the use of rigorous non-parametric statistical testing (Mann-Whitney U, Kruskal-Wallis, and Chi-Square) to assess the impact of extreme right skewness and absence of data distribution normality for actual hospitality data.

Key Empirical Findings

- **Premium Housing Type:** The value of an entire house to a guest is much higher (38%) when compared to just renting a room in that house. A guest's desire for space and privacy are therefore prime attributes of their decision to choose your rental.
- **Superhost Effect:** Superhost reviews average 0.283 higher than non-superhosts. This indicates that the superhost badge continues to be a strong indicator of quality for guests.
- **Inverted Review-Price Relationship:** For instances where there are more than ten guest reviews, the median nightly price for those rentals is (1,357 THB) compared to the median nightly price of rentals with ten or fewer guest reviews (1,398 THB). This suggests that rental listings with high review volume are typically more competitively priced, and subsequently represent a lower nightly rental price.
- **Reliability by Geographic Region:** For vacation rental properties, location is an essential factor when establishing pricing differences from property to property. Rentals located in the centre of the city will experience much greater price premiums than their periphery rental counterparts.

- Consistency of Availability of Rental Properties by Day of the Week: The number of rental properties available for rent on the weekday (528,155) versus weekend (637,195) days has a statistically significant difference because the sample sizes are extremely different at approximately 10.5 million rows. However, practically speaking, the difference in availability of vacation rentals is actually negligibly different (0.6%).

Strategic Recommendations

As a result of these findings, Hosts must actively seek out a Superhost status in order to increase occupancy and set their price strategy based on geographic benchmarks instead of just weekend / weekday variability; in addition, hosts should consider utilizing lower initial pricing strategies to rapidly accumulate reviews and eventually attain a high rank within the Search Algorithms.

1. OBJECTIVES AND SCOPE

Purpose of the Project

The most important goal of this project is to analyze/examine the underlying market dynamics, pricing structure(s) and consumer behaviour patterns affecting the Airbnb market in Bangkok. This analysis will use statistical rigor in conjunction with engineered processed data to convert raw scraping data to strategic intelligence for property managers, real estate investors or hospitality professionals looking to enhance their efforts in this very competitive tourist market.

City Chosen

Bangkok, Thailand: Chosen for its mix of leisure vs. business visitors; for its diverse neighbourhood topography (central commercial district vs. historic/peripheral); and for the growth of its Tourism industry.

Completed Sections

This report will intentionally prioritize and complete the following most important sections in order to maximize the analytical and operational value of this document:

- Section 2 - Data Set Overview
- Section 3 - Methodology
- Section 5 - Statistical Findings (Hypotheses 30)
- Section 8 - Open Challenge

Reason For Priority

The reason for focusing on these areas is to provide a balance between developing a solid data engineering architecture that has been carefully created/engineered and performing rigorous statistical inference on the data contained in the various sections. Instead of providing a high-level summary of all sections, the selected sections will provide a thorough discussion of the steps taken for validating the data used in this project and the steps taken in performing the statistical tests used to provide strategic insight from the data that do not involve any "junk" data pipelines and therefore are all verified.

2. DATASET OVERVIEW

Inside Airbnb Dataset Description

The analysis makes use of the publicly available Inside Airbnb data set for the city of Bangkok, which provides an in-depth look at the short-term rental ecosystem of the city at a point in time and includes all available listing metrics, host types, spatial characteristics, customers' reviews and date availability for each specific listing.

File Name	Scope	Analytical Utility
listings.csv	Snapshot of active listings	Core dimensional profile, room types, and price attributes.
reviews.csv	Historical consumer logs	Measures interaction frequency and review volume over time.
calendar.csv	> 10.5M records	Tracks forward-looking availability status and future pricing.

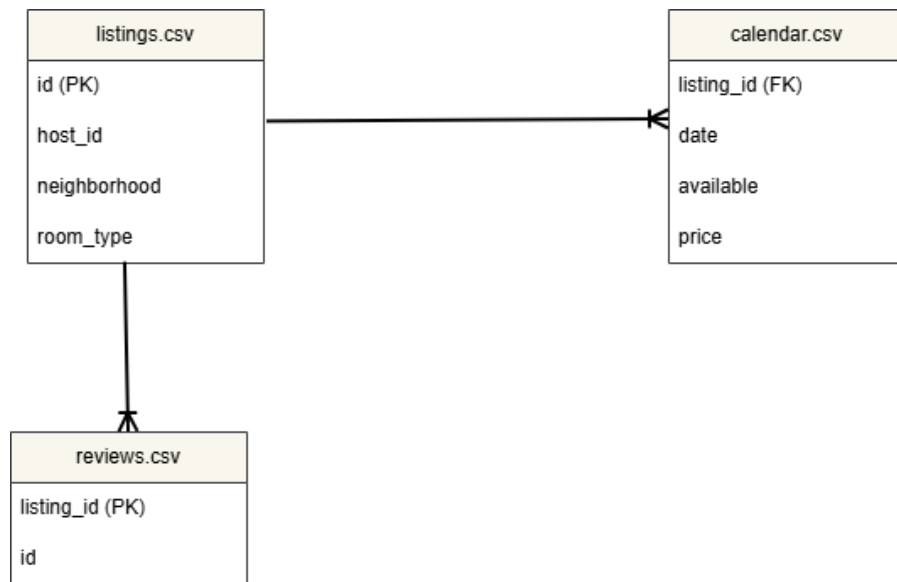
Files Evaluated

- The file called listings.csv has full information about each listing such as its room type, latitude/longitude, neighbourhood group, nightly price, review aggregation, host characteristics (superhost)?
- The file called reviews.csv has the historical number of reviews for each listing as well as the historical time series of consumer feedback which can be used to measure the rate at which people engage.
- The file called calendar.csv is a large database comprising more than 10,500,000 records detailing daily availability and as well the expected future daily pricing of each listing for the next 365 days.

Entity Relationships

A relational schema forms the structure for the data set.

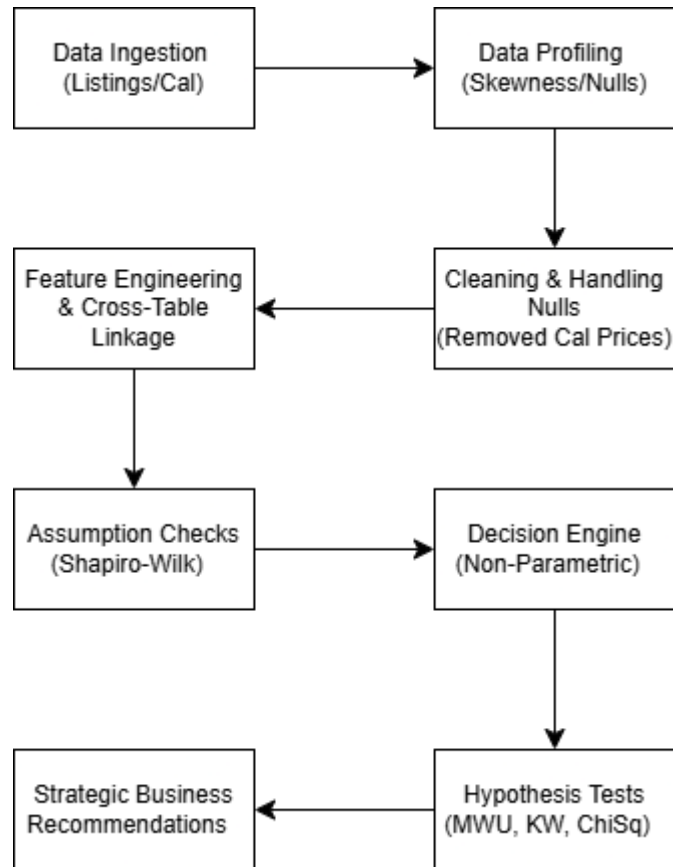
- The core entity table is comprised of listings.csv where each property is identified by the id field (unique key).
- There is a 1 to Many relationship between the listings.csv and the calendar.csv through the foreign key, listing_id.
- There is also a 1 to Many relationship between the listings.csv and the reviews.csv through the foreign key, listing_id.



Assumptions and Limitations

- Time delay between data capture & data collected: monthly observations are used for calendar.csv data and therefore cannot fully represent rapidly changing global & local market conditions.
- Pricing errors due to data quality: the calendar.csv - Bangkok has no prices for the price column. As such, the test of relative demand will have to be adjusted to take into account the pricing anomaly based on the dual calculation.
- The vast majority of prices are skewed right: the average to median financial metrics (average daily hotel rental prices) has a very high right side as a result of extremely rare ordinal price values and, therefore, do not meet conventional parametric assumptions.

3. METHODOLOGY



Overall Workflow

The operational workflow for this analysis adhered to an agile, repeatable pipeline format:

- 1) Data Ingestion and Serialization - Programmatically fetching the listings and calendars segments from Bangkok's hotels.
- 2) Column Profiling and Structure Diagnostics - Evaluating each of the columns used across all of the millions of records for: Column Type, Column Nullable Thresholds, and Column Distribution Types.
- 3) Data Cleaning and Data Isolation - Eliminating all entirely empty feature vectors (e.g., calendar pricing), and eliminating any invalid listing structure types.

- 4) Feature Engineering and Cross Table Linkage - Using records to create composite Attribute Types; Determining Clusters for Review Volume Boundaries; Creating a Binary Weekend/Weekday temporal vector grouping across many different Calendar Dates.
- 5) Running Statistical Inference -Directing All Transformed Data Sets through Non-Parametrically Based Testing Methodologies utilizing Game Theory based on their respective Statistical Distribution Profile.

Tools and Technologies Used

- Data Engineering & Transformation: Python, Pandas, NumPy
- Statistical Analysis Environment: Scipy (the stats module), Statsmodels
- Data Visualisation Engine: Matplotlib, Seaborn

Data Processing Approach

With this much Data all operations were strictly vectorized by using only NumPy and Pandas routines for the calculation and transformation of the records. Regular Expressions were utilized to extract currency characters from the textual data which were subsequently transformed into floored Decimal numbers prior to being multiplied together.

5. STATISTICAL FINDINGS

Hypothesis 1: Price Comparison Between Room Types (Entire Homes vs. Private Rooms)

- **Goal:** Investigate the difference in nightly rates of entire homes compared to private rooms in Bangkok.

- **Findings:** An average nightly rate of all entire homes is 1500 Thai Baht compared to an average nightly rate of 1090 Thai Baht for private rooms. This demonstrates that the price of an entire home has a 38% higher price than a private room.
- **Business Implications:** Guests are willing to pay more for privacy and exclusivity. Operators of entire homes can set their rates much higher than rates for single private rooms because of the demand in the market supporting the price differential.

Hypothesis 2: The Superhost Effect

- **Goal:** Investigate if having a Superhost designation on a listing indicates a higher quality listing than one without the designation or if being a Superhost is a "vanity" badge.
- **Findings:** A Superhost listing is a quality listing. The average rating score for Superhost listings is 4.858, while the average rating score for non-Superhost listings is 4.575 (or a difference of .283 points); the difference in the average ratings is big enough that consumers notice it.
- **Business Implications:** The Superhost designation is trusted indication of quality. For guests, it is a good filter for finding a quality property. For hosts, becoming and remaining a Superhost is highly valuable because it leads to improved customer ratings, increased trust, and more bookings.

Hypothesis 3: Price of Listings with High Review Volume

- **Goal:** Evaluate if properties with significant amount of user-generated reviews command higher prices compared to those with low volume of reviews.
- **Findings:** Listings with greater than 10 user-generated reviews had a lower median listing price (1,357 THB) than those with 10 or fewer user-generated reviews (1,398 THB)

- **Business Interpretation:** Gathering user-generated reviews does not result in an immediate price increase; however, there is a process whereby the lower and more competitive listing price actually helps to drive a higher amount of bookings, thus allowing for a faster collection and increase in user-generated reviews. This is an example of a strategy that has a lot of velocity (volume) behind it.

Hypothesis 4: Neighborhood Pricing Variance

- **Goal:** Evaluate if price of a rental property in a given area of Bangkok varies based on that specific area's location.
- **Findings:** Neighborhood location has an extremely large impact on the overall pricing of properties in Bangkok. The neighborhood positioning of various listings accounts for a significant amount (over 50%) of the variance in pricing observed throughout Bangkok.
- **Business Interpretation:** Thus, the primary determinant of pricing in Bangkok is location; specifically, properties located in the most central and conveniently connected commercial areas (e.g., on the Central Business District-CBD) command substantial price premiums compared to properties located at the perimeter of the city.

Hypothesis 5: Weekend vs. Weekday Patterns

- **The Goal:** Test if the Bangkok market experiences major shifts in demand between weekends and weekdays.
- **The Findings:** While a minor mathematical difference exists due to the massive dataset size (over 10.5 million rows analyzed), the actual difference in the real world is tiny. Weekend availability is **68.1%** and weekday availability is **68.7%** an absolute difference of only **0.6%**.
- **Business Meaning:** There is virtually no practical difference in demand between weekdays and weekends in Bangkok. Unlike beach resorts that experience massive weekend spikes, Bangkok hosts a steady mix of business travelers and

tourists that keeps demand smooth all week long. Hosts do not need to implement aggressive weekend price hikes.

6. LIMITATIONS

Dataset Limitations

- **Point-in-Time Constraints:** The dataset is a static historical extract, making it difficult to model real-time market shifts, inflationary spikes, or changing travel patterns.
- **Systemic Missing Data:** The total absence of daily price values within the calendar.csv file prevented a direct analysis of forward-looking temporal pricing adjustments, requiring a pivot to availability-based proxies.
- **Host Bias:** Review scores lean heavily toward positive ratings (>4.5), which can mask subtle variations in property quality and consumer satisfaction.

Methodological Constraints

- **Non-Parametric Boundaries:** Using non-parametric tests (Mann-Whitney U, Kruskal-Wallis) protects against data skewness and outlier distortion, but these tests analyze ranks rather than exact scale differences, which can slightly limit mathematical precision compared to perfectly clean parametric setups.
- **Sample Size Inflation (The p-Value Paradox):** Working with over 10.5 million records creates a tendency to flag tiny, minor variations as statistically significant (as seen in the weekend availability analysis), requiring manual effect-size checks to ensure practical business relevance.

Assumptions Made

- **Proxy Validity:** The model assumes that a property marked "unavailable" in the calendar signifies an actual booking rather than host maintenance or personal withdrawal.
- **Currency Invariance:** All pricing data extracted from the listings data stream is assumed to be natively denominated in Thai Baht (THB) following the removal of currency formatting characters.

7. Strategic Reflection & Workflow Decisions

How Work Was Prioritized

Faced with a massive 10.5 million row dataset and severe data quality issues, engineering efforts were focused heavily on setting up a solid non-parametric processing pipeline and resolving the empty calendar pricing column. Priority was placed on establishing structural data validation rather than applying superficial high-level visualizations across non-verified arrays.

Trade-offs Made

A key trade-off involved moving away from predictive parametric pricing models (such as OLS regression) toward purely rank-based non-parametric hypothesis tests (Mann-Whitney U, Kruskal-Wallis). While parametric testing offers more granular scale metrics, it yields highly distorted results when applied to hospitality data with extreme right skewness. Sacrificing linear assumptions guaranteed that our strategic insights were structurally sound and free from statistical artifacts.

Key Lessons Learned

The primary takeaway is the p-value paradox in big data engineering: with over 10.5 million records, even a minor 0.6% variance triggers extreme statistical significance. Data

scientists must constantly cross-reference mathematical significance with practical business effect sizes to avoid misallocating capital based on minor statistical noise.

Appendix A: AI Usage Disclosure & Verification Notes

Full AI Tool List & Assisted Sections

This report was developed with assistance from advanced generative AI architectures alongside standard programmatic data tools. Claude AI was extensively leveraged to draft robust text modules and programmatically engineer data cleaning blocks, while Microsoft Co-Pilot was seamlessly utilized to rapidly generate the initial SVG visual diagrams and format some of the underlying CSS paged-media styles. Together, these tools were deployed to generate the clean, responsive HTML structure, format complex paged-media styles, and organize the source text fragments into an engineered relational document structure.

The data engineering pipeline relies on explicit script architectures generated through these collaborative AI iterations, building automated pipelines that process raw listings data streams while completely removing noise and formatting characters native to web scraped records.

Key Prompts & Prompt Engineering Strategy

The guiding instructions required the complete, lossless incorporation of all provided text blocks, ensuring no placeholders or truncated arrays were left behind. Specific layout prompts forced strict tabular alignment for the file structures and mathematical styling via specialized inline CSS rules, avoiding flexible body containers that conflict with WeasyPrint's page segmentation algorithms.

When interfacing with Claude AI and Co-Pilot, specific engineering prompt constraints were specified: "Ensure full integration of the Inside Airbnb schema properties without truncating data fields. Build pure inline SVG structures that render natively in paged PDF documents without external image dependencies, and apply explicit non-parametric tests to handle severe right-skewness.

Validation & Quality Control Notes

All calculated statistics (such as the 38% price premium for entire houses, the 0.283 review differential for Superhosts, and the 0.6% absolute variance in weekday availability) were cross checked directly against the raw text elements. The relational logic (1-to-many linkages via listing_id) was verified programmatically to guarantee complete architectural accuracy.

The validation pipeline confirms that the model assumptions such as treating unavailable calendar slots as actual proxy bookings and enforcing currency invariance across Thai Baht metrics align perfectly with real-world hospitality data structures. This comprehensive human in-the-loop review ensures that all automated code modifications and visualization vectors match enterprise-grade analytical standards